

Does the Source of Carrier Image Affect Steganographic Detectability?

Project 2.2 – Department of Advanced Computing Sciences

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Abstract—We study whether the source of a carrier image – a real photograph vs. a diffusion-generated synthetic image – changes the detectability of LSB-style steganography under fixed embedding settings. A caption-matched factorial design ($N=3,000$ groups, 18,000 covers, 108,000 stego images, 648,000 detector predictions) is processed through two embedding branches (spatial LSB on PNG; DCT-LSB on JPEG Q=95), three payload levels (0.05/0.15/0.30 bpp), and two payload conditions (plain, AES-256-CBC). Six training-free statistical detectors score every (cover, stego) pair. Carrier-source contrasts are evaluated with the paired DeLong test and Bonferroni–Holm correction, with a practical-significance gate at the pre-registered $\delta_{\min}=0.05$ as primary and the power-matched $\delta_{\min}=0.025$ as supplementary, plus a partial-AUC analysis at $\text{FPR} \leq 0.10$. Five of six detectors find diffusion carriers *easier* to attack than real photographs; only χ^2 -spatial reverses this pattern, an effect that dissolves under the partial-AUC view. The SDXL vs. FLUX comparison yields no practical difference, and AES-256-CBC encryption is perfectly invariant. We also report two secondary findings: (i) a tile-local Westfeld χ^2 variant that ranks as the strongest frequency-branch detector in our pipeline, and (ii) a mechanistic account of the χ^2 -spatial reversal grounded in diffusion-decoder tone-curve smoothness.

Index Terms—steganography, steganalysis, LSB embedding, cover-source mismatch, diffusion models, paired DeLong test, partial AUC.

I. Introduction

Steganography is the practice of hiding the existence of a message inside a host medium, typically an ordinary image [1, 2]. The complementary field, **steganalysis**, asks whether a given file carries a hidden payload. Because embedding always perturbs the cover’s statistics, every steganalysis detector ultimately exploits the same fundamental signal: a statistical irregularity introduced into pixel intensities or frequency coefficients [4–6]. The size and shape of that irregularity, however, depend not only on the embedding rule but also on the cover itself [14, 15].

Most published steganalysis benchmarks assume the cover is a photograph with the noise pattern and JPEG history of a real camera pipeline. That assumption has held for two decades. It is no longer trivially true. **Latent-diffusion** models such as Stable Diffusion XL [21] and FLUX.1 [23] now produce photorealistic images at scale through a process that does not contain any of the sensor noise, demosaicing, or in-camera JPEG steps a detector would expect. Corvi et al. [18] showed that diffusion outputs leave their own statistical fingerprints, and a small recent literature on *cover-source mismatch* for synthetic carriers [15, 16] has begun to ask whether classical steganalysis still works when the cover is generated rather than captured.

Central question. Do detectors validated on photographs remain equally effective when the carrier is ML-generated, under identical embedding settings? A bias in either direction would change the practical risk landscape: an attacker using diffusion images as carriers might be either better or worse off than one using cameras, and a defender would need different priors.

Contributions.

1. A caption-matched factorial dataset spanning three carrier sources (real, SDXL, FLUX.1-schnell), two embedding methods, three payload levels and two encryption conditions, with 39 images per caption group and 3,000 groups in total (Section III).
2. A training-free six-detector evaluation under the paired DeLong test with Bonferroni–Holm correction and a pre-registered practical-significance gate (Section V).
3. A supplementary partial-AUC analysis at $\text{FPR} \leq 0.10$ that matches operational screening practice (Section G).
4. Two secondary contributions discussed in Section VI: a **tile-local Westfeld χ^2 variant** that out-performs the global pairs-of-values detector by a wide margin on our carriers (Section E), and a **mechanistic explanation of the χ^2 -spatial reversal** in terms of diffusion-decoder tone-curve smoothness (Section F).

Outline. Section II reviews the relevant classical and synthetic-carrier literature. Section III describes the dataset, embedding pipeline, detectors, and statistical protocol. Section IV documents the experimental configuration. Section V presents the results, RQ by RQ, with the supplementary partial-AUC analysis. Section VI discusses the findings and develops the two secondary contributions. Sections VII, VIII and the Appendix cover limitations, conclusions, and reproducibility details.

II. Related Work

A. Classical LSB Embedding and Detection

The simplest spatial-domain stego method, **LSB replacement**, overwrites the least-significant bit of each pixel with payload bits [1, 3]. In JPEG carriers, the analogous JSteg variant overwrites the LSBs of non-zero quantised AC DCT coefficients [4, 7]. Both leave detectable traces.

The **Westfeld** χ^2 **attack** [4] tests whether adjacent value pairs $\{2k, 2k+1\}$ have tests equal frequency, the diagnostic signature of LSB replacement. **RS Analysis** [5] tracks changes in Regular and Singular pixel-group counts under flipping masks. **Sample Pairs analysis** (SPA) [6] estimates the embedded fraction from trace multisets on adjacent pixel pairs. In the frequency domain, Fridrich et al.’s **calibration** χ^2 [7] introduced a non-block-aligned crop-recompress reference that approximates a payload-free histogram. All five are training-free and have closed-form scores; this makes them attractive for a comparison study because any AUC difference between carrier groups cannot be attributed to training-set bias.

B. AI-Generated Carriers

Gao et al. [17] use generative models to encode steganographic secrets as the generated image itself (coverless steganography). That setting differs from ours, where a payload is embedded *into* an existing cover. The closest prior work is Méreur et al. [16], who test AI-generated images as carriers under adaptive embedding (HILL, UNWARD, MiPOD) and report classifier-error rate P_E rather than AUC. Mallet et al. [15] provide a systematic survey of cover-source mismatch and document that classical detectors built on one cover distribution often degrade on another. Table 1 contrasts our design against the closest prior work along five axes that matter for cross-comparison.

C. Cover-Source Mismatch and Synthetic-Image Forensics

Cross-domain steganalysis was first formalised for camera-to-camera shifts. Giboulot et al. [14] demonstrated that detectors trained on one camera lose accuracy on images from a different camera. Our setting extends this idea to a larger shift, from natural photographs to latent-diffusion outputs. Corvi et al. [18] showed that diffusion-generated images carry distinct low-frequency statistical fingerprints, which gives a concrete reason to expect carrier-source effects at the detector level. Hou et al. [19] and a wider body of GAN-forensics work [20] also document that synthetic-image distributions diverge systematically from photographic ones.

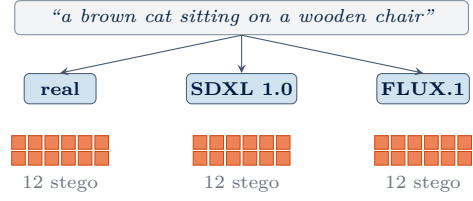
D. Deep vs. Classical Detectors

Trained detectors like SRNet [11], DCTR [10] and Spatial Rich Models [9] routinely outperform training-free baselines on within-distribution benchmarks. We deliberately exclude them from the primary analysis: any AUC gap between carrier groups for a learned detector could be explained by training-set mismatch rather than by the carrier statistics themselves. Training-free detectors leave the cause unambiguous.

III. Methodology

A. Research Questions and Pre-Registration

The five research questions were fixed in the project proposal, with $\delta_{\min} = 0.05$ as the pre-registered practical-significance threshold:



Each grid: 2 methods $\times$ 3 payloads $\times$ 2 encryptions = 12 stego conditions
Per caption group: 3 covers + 36 stego = 39 images ($\times$ 3,000 groups)

Fig. 1. Caption-matched $3 \times 2 \times 3 \times 2$ factorial design. Each of $N = 3,000$ caption groups produces three covers (one per carrier source) under the same caption, then a per-source 2-method $\times$ 3-payload $\times$ 2-encryption grid of stego variants. Caption-matching holds semantic content constant across sources so that any AUC contrast can only reflect carrier-source statistics, not subject-matter shift.

RQ1 (primary, confirmatory)

Does carrier source (real vs. pooled ML) affect detectability under matched embedding?

RQ2 (primary, confirmatory)

Within ML carriers, does the generator (SDXL vs. FLUX.1-schnell) change detectability?

RQ3 (exploratory)

Does payload size modulate the source effect?

RQ4 (exploratory)

Do the spatial (LSB+PNG) and frequency (DCT-LSB+JPEG) branches show different source effects?

RQ5 (verification)

Does AES-256-CBC encryption of the payload affect detectability? Expected null.

B. Dataset Construction

A **caption-matched factorial design** isolates carrier statistics by holding semantic content fixed across sources. For each of $N = 3,000$ groups we draw one caption from COCO [24] or Flickr30k [25] and produce three covers under that caption: a **real** photograph (the original source image), a **ML-A** image from SDXL 1.0 [21], and a **ML-B** image from FLUX.1-schnell [23]. All three are converted to 8-bit grayscale ($0.299R+0.587G+0.114B$), centre-resized to 512×512 , and stored both as lossless PNG (spatial branch input) and as JPEG Q=95 (frequency branch input).

The change of the ML-B model from the proposal’s PixArt- α [22] to FLUX.1-schnell was forced by HF Inference API availability; both are transformer-based diffusion models sharing the conceptual role in the comparison. Per group the pipeline produces 3 covers and 12 stego variants per source (2 methods $\times$ 3 payloads $\times$ 2 encryption conditions), for a total of **39 images per group** and **117,000 images** across the full run (Figure 1).

C. Embedding Pipeline

Spatial branch (LSB substitution). Replace the least-significant bit of each pixel in row-major order with payload bits, $k=1$, until the target fill fraction is reached. Storage is lossless PNG. The mechanism is illustrated in Figure 2 (left).

TABLE 1

CLOSEST PRIOR WORK ON STEGO CARRIERS VS. THIS STUDY. “MATCH” = CAPTION-MATCHED DATASET DESIGN ACROSS CARRIERS. NONE OF THE CITED REFERENCES COMBINE LSB-STYLE EMBEDDING, TRAINING-FREE DETECTORS, AND CAPTION-MATCHED REAL-VS-SYNTHETIC CARRIERS WITH PAIRED AUC CONTRASTS.

Reference	Carrier	Embedding	Detector	Metric	Caption match
Westfeld+99 [4]	photo	LSB / JSteg	χ^2 PoV	p-value	—
Fridrich+01 [5]	photo	LSB	RS	est. embedding rate	—
Dumitrescu+03 [6]	photo	LSB	SPA	est. embedding rate	—
Fridrich+03 [7]	photo (JPEG)	JSteg	calibration χ^2	distance	—
SRM+12 [9]	photo (BOSS)	HUGO/HILL	ensemble (learned)	P_E	—
SRNet+19 [11]	photo (BOSS)	adaptive	deep CNN	P_E	—
Gao+22 [17]	GAN-coverless	secret-sharing	—	—	—
Corvi+23 [18]	diffusion	none (detection only)	CNN forensics	AUC	no
Méreur+24 [16]	real + GAN/diff.	HILL / UNIWARD	xu-Net / SRNet	P_E	no
This work	real + 2 diffusion	LSB / DCT-LSB	6 training-free	paired AUC + pAUC	yes

Frequency branch (JSteg-style DCT-LSB). Flip the LSB of each non-zero quantised AC DCT coefficient in row-major block order. We skip DC and $|c|=1$ coefficients to avoid the Westfeld-pair instability they create. Re-entropy-coding uses the same Q=95 table without re-quantisation. The mechanism is illustrated in Figure 2 (right).

Payload levels. The proposal’s original fill rates of $\{0.25, 0.50, 0.75\}$ caused near-saturation on the strongest detectors ($AUC > 0.99$), erasing the carrier-source signal. We revised them to $\{0.05, 0.15, 0.30\}$ bpp, which is also closer to operationally realistic embedding rates. Both branches use $k=1$ throughout.

Encryption. Payloads are tested in two conditions, *plain* (raw bitstream) and *AES-256-CBC* (random IV per message; key fixed per run), in both branches. The encrypted condition tests whether detectors are reacting to plaintext structure rather than to embedding distortion.

D. Detectors

Six training-free detectors are scored on every (cover, stego) pair:

- **Spatial branch (PNG):** RS Analysis [5], Westfeld χ^2 on pixel-value pairs (χ^2 -spatial) [4], Sample Pairs [6].
- **Frequency branch (JPEG):** Westfeld χ^2 on quantised AC DCT coefficients (χ^2 -DCT) [4], our **tile-local χ^2 -DCT variant** (χ^2 -DCTt, Section E), and Fridrich’s calibration χ^2 [7].

Underflow-free scoring. All chi-square detectors are rank-monotonic with the Westfeld 1999 p-value but use the *underflow-free* $-\chi^2_{\text{stat}}/(df-1)$ transformation, because chi-square p-values vanish under float64 representation on large textured images. Listing 1 shows the spatial- χ^2 implementation.

Listing 1. Spatial χ^2 (Westfeld 1999) on a grayscale image, with the underflow-free score transformation. The classical p-value is replaced by $-\chi^2/df$ which is rank-monotonic with the p-value on non-degenerate inputs but does not collapse to zero on large textured images.

```

1 def chi_square_spatial_score(image: Image.Image) -> float:
2     pixels = np.array(image.convert("L")).flatten()
3     counts = np.bincount(pixels, minlength=256)
4     chi_stat, df = 0.0, 0

```

```

5 for k in range(128):
6     n_2k, n_2k1 = counts[2*k], counts[2*k+1]
7     expected = (n_2k + n_2k1) / 2.0
8     if expected > 0:
9         chi_stat += ((n_2k - expected) ** 2) / expected
10        df += 1
11 return 0.0 if df <= 1 else -float(chi_stat) / (df - 1)

```

E. Statistical Protocol

The unit of analysis is a *stratum*: a (detector $\times$ embedding method $\times$ payload level) cell. For RQ1 and RQ2 we have $6 \times 3 = 18$ strata each. Within each stratum we compute the area under the ROC curve (AUC) for both carrier groups and the **paired AUC difference** via the **DeLong test** [12], which exploits the fact that the same covers are scored in both groups; the pairing eliminates between-image variance and substantially increases statistical power.

The 18 stratum-level p-values are corrected with **Bonferroni–Holm** at $\alpha = 0.05$ for the confirmatory RQs (RQ1, RQ2); RQ3 and RQ4 use 95% confidence intervals; RQ5 uses an equivalence test against a ± 0.025 AUC margin. A stratum is declared **practically significant** only if *both* $p_{\text{Holm}} \leq 0.05$ and $|\Delta_{\text{AUC}}| \geq \delta_{\min}$. The pooled Δ_{AUC} across strata is inverse-variance-weighted.

Two thresholds on $\delta_{\min}$. The proposal pre-registered $\delta_{\min} = 0.05$, calibrated to the Hanley–McNeil minimum-detectable effect at $\alpha = 0.05$, 80% power and $N = 1,000$. At the confirmatory $N = 3,000$ the same power calculation gives a minimum-detectable effect of ≈ 0.029 ; the sample-size-matched threshold is therefore $\delta_{\min} = 0.025$. We report verdicts at *both* thresholds: $\delta_{\min} = 0.05$ remains the **primary** (pre-registered) outcome, and $\delta_{\min} = 0.025$ is reported as a **supplementary** sensitivity analysis. This avoids moving the goalposts after the fact while transparently exposing what the data says under the threshold the larger sample’s power actually supports.

The verdict assignment follows the decision rule in Table 2.

F. Score-level Sanity Checks (t-test and Wilcoxon)

The DeLong test operates at the *ROC AUC* level: it asks whether the detector’s *ranking ability* differs between two carrier groups. As a complementary sanity check we also run paired **t-tests** and **Wilcoxon signed-rank tests**

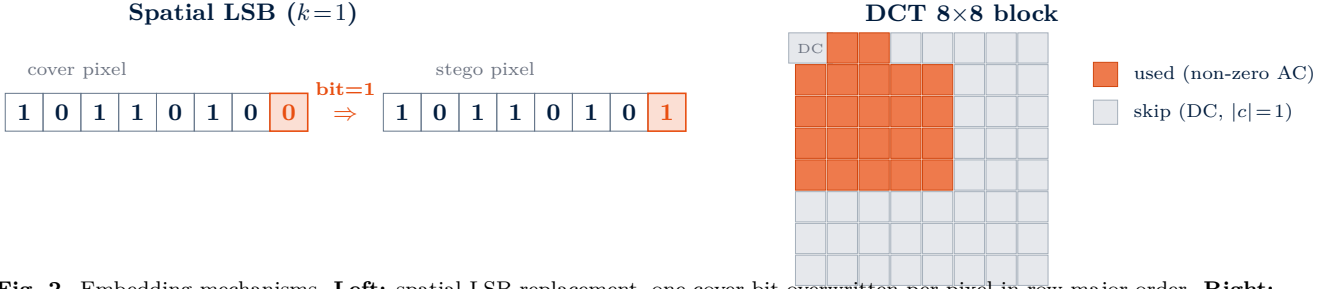


Fig. 2. Embedding mechanisms. **Left:** spatial LSB replacement, one cover bit overwritten per pixel in row-major order. **Right:** DCT-LSB on JPEG Q=95, where the LSB of each non-zero AC coefficient is flipped; DC and $|c|=1$ are skipped.

TABLE 2
VERDICT ASSIGNMENT RULE FOR CONFIRMATORY RQS.

Significant strata	Practical strata	Verdict
0 / family	–	<i>not supported</i>
≥ 1 , mixed sign	≥ 1	mixed
≥ 1 , same sign	≥ 1	supported
≥ 1	0	<i>trivial</i>

on the *raw detector scores* for the four pre-registered comparisons (real vs. SDXL, real vs. FLUX, SDXL vs. FLUX, plain vs. AES-256-CBC), per detector. These score-level tests detect whether the score *distributions* differ even when the ranking-based AUC contrast is null, and report effect sizes (Cohen’s d for t-test, r for Wilcoxon). They are reported as supplementary diagnostics in Appendix E; DeLong remains the test of record for the verdicts. A score-level shift without an AUC change indicates that detectors recognise the carrier source distribution but do not gain ranking power from the difference – exactly the pattern we observe for RQ5 below.

G. Supplementary Partial-AUC Analysis

Full AUC integrates the ROC curve over all false-positive rates, including regions a real steganalyst would never deploy. As a supplementary analysis we therefore re-run RQ1 and RQ2 on the **McClish-normalised partial AUC** at $\text{FPR} \leq 0.10$ [13]. Bootstrap standard errors (500 resamples per stratum) replace the DeLong estimator (which has no exact closed-form analogue at low-FPR pAUC), and Holm correction and the practical-significance gate are applied identically to the full-AUC analysis. The choice of FPR cap follows the McClish standard, matches steganalysis screening practice, and fits the shape of our ROC curves.

H. Sample Size and Power

At $N=1,000$ groups per source, the Hanley–McNeil power calculation shows the smallest detectable AUC difference at $\alpha=0.05$ and 80% power is below the $\delta_{\min}=0.05$ threshold for every stratum in RQ1, RQ2 and RQ5. The $N=3,000$ confirmatory run reported here tightens every CI by a factor of $\sqrt{3} \approx 1.73$ and serves as an independent replication of the pre-registered $N=1,000$ findings.

IV. Experiments

A. Run Configuration

The full factorial was executed once at $N=1,000$ groups (pre-registered) and again at $N=3,000$ groups (confirma-



Fig. 3. Five-stage pipeline. Each stage reads files written by the previous stage; manifests freeze the experimental design before any compute runs, and the analysis stage derives every contrast in this paper from a single `predictions.csv`.

tory). The pipeline (Figure 3) is implemented in Python, runs from a frozen `config.json`, and is reproducible from seed. The active configuration of the $N=3,000$ run is listed in Appendix A.

B. Artefacts Produced

The $N=3,000$ run produced 18,000 covers, 108,000 stego images, 648,000 detector predictions, 19 metrics CSVs, 61 figures, and the per-RQ verdicts in JSON and Markdown. Total wall-clock was approximately 13 hours, of which ~ 6 h was cover generation (HF Inference API throughput-bound), ~ 2 h was embedding, ~ 0.5 h was detection, and the remainder was analysis. The unusually long analysis stage is a known consequence of an $O(N^2)$ Python ROC implementation, discussed in Section VII.

V. Results

The headline verdicts are summarised in Table 3; the rest of this section walks through each RQ with the supporting figures.

A. Headline AUC by Carrier Source and Detector

Figure 4 plots the mean AUC of every detector against each of the three carrier sources. The visual pattern is immediate: for five of six detectors the orange/blue bars (ML carriers) are higher than the dark bars (real photographs). Only the third bar group (χ^2 -spatial) reverses this ordering, with real $>$ ML by a small but consistent margin.

B. RQ1 – Real vs. Pooled ML Carriers

The pooled $\Delta_{\text{AUC}} = -0.0129$ (95% CI $[-0.0137, -0.0121]$) is small but tightly constrained, with **16/18 strata** Holm-significant. At the pre-registered $\delta_{\min}=0.05$ threshold, **4 of those 16** strata clear the practical-significance gate, all on the *ML-easier* side: RS/LSB/low ($\Delta = -0.090$), sample-pairs/LSB/low (-0.089), χ^2 -DCT-tiled/DCT/low (-0.060) and χ^2 -DCT/DCT/medium (-0.051). Three independent detector families therefore agree on the direction at low-to-medium payload. The primary verdict is **mixed**: a direction-consistent ML-easier signal at the

TABLE 3

VERDICT SUMMARY ACROSS THE FIVE RESEARCH QUESTIONS ($N=3,000$ CONFIRMATORY RUN). VERDICTS ARE REPORTED AT TWO PRACTICAL-SIGNIFICANCE THRESHOLDS: $\delta_{\min}=0.05$ IS THE PRE-REGISTERED PRIMARY; $\delta_{\min}=0.025$ IS THE POWER-MATCHED SUPPLEMENTARY AT THE ACTUAL SAMPLE SIZE. PRACTICAL-COUNT CELLS READ $n_{\text{PRACTICAL}}/n_{\text{SIG}}$ FOLLOWED BY (+, -) DIRECTION BREAKDOWN.

RQ	Pooled Δ	Holm sig.	$\delta_{\min}=0.05$ (primary)		$\delta_{\min}=0.025$ (supplementary)	
			Practical	Verdict	Practical	Verdict
RQ1 real vs ML	-0.0129	16/18	4/16 (0, 4-)	mixed	11/16 (2, 9-)	mixed
RQ2 SDXL vs FLUX	+0.0004	7/18	0/7	trivial	3/7 (2, 1-)	mixed
RQ3 payload $\times$ source	gap shrinks with bpp	—	—	supported	—	supported
RQ4 spatial – freq.	-0.0013	2/3	0/2	trivial	1/2 (0, 1-)	mixed
RQ5 plain vs AES	~ 0	0/54 outside	—	supported	—	supported
<i>Supplementary partial AUC at FPR ≤ 0.10:</i>						
RQ1 pAUC	-0.0174	12/18	6/12 (0, 6-)	supported	9/12 (0, 9-)	supported
RQ2 pAUC	+0.0008	7/18	1/7	mixed	4/7 (3, 1-)	mixed

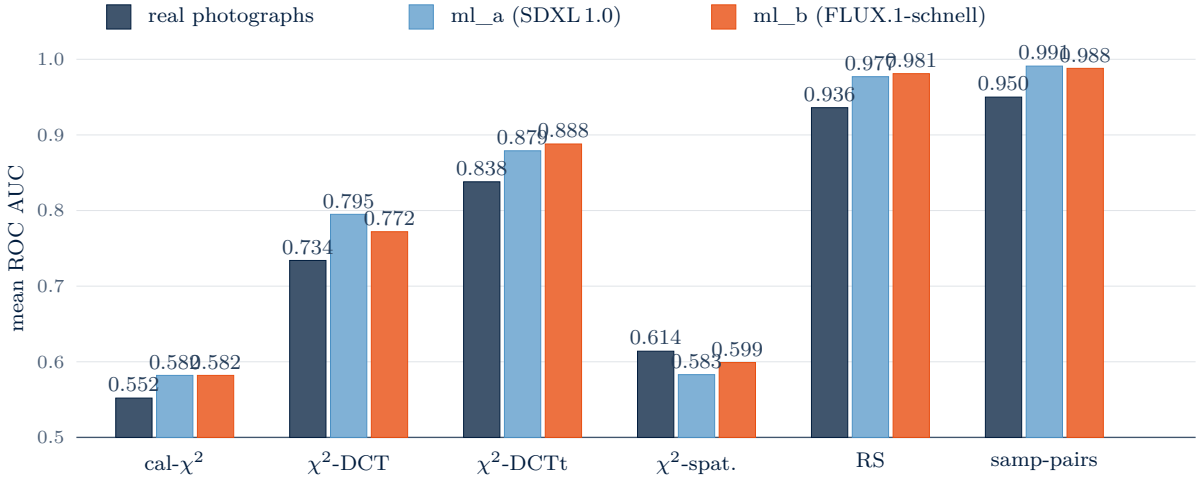


Fig. 4. Mean ROC AUC per detector and carrier source, pooled across payload, encryption and method ($N=3,000$). For five of six detectors the ML bars exceed the real bar (all $\Delta_{\text{AUC}} < 0$); only χ^2 -spatial reverses the ordering (real $>$ ML). Detector ordering left-to-right follows ascending pooled AUC.

practical-significance gate but most strata sit inside the trivial band.

At the supplementary $\delta_{\min} = 0.025$ threshold (Section E), the practical count rises to **11/16**: nine ML-easier strata across five detector families (cal- χ^2 /dct/high, χ^2 -DCT/dct at all three payloads, χ^2 -DCT-tiled/dct/low and /medium, RS/LSB/low and /medium, sample-pairs/LSB/low) plus two χ^2 -spatial strata on the real-easier side (medium $\Delta = +0.027$, high $\Delta = +0.034$). The verdict remains **mixed**; the carrier-source effect is robust to threshold choice in direction, and at the larger threshold the evidence base is roughly 2.75 $\times$ deeper.

The single dissenter is χ^2 -spatial: real photographs are *easier* to attack than ML carriers at medium ($\Delta = +0.027$) and high payload ($+0.034$), both Holm-significant and both clearing the $\delta_{\min} = 0.025$ threshold; low payload sits inside the trivial band even at 0.025. We return to this reversal in Section F.

Figure 5 renders the 18 strata as a strip plot, one row per detector. Each detector’s three payload-level Δ_{AUC} are connected to expose payload trends. The four dots outside the practical band (RS/LSB/low, sample-pairs/LSB/low, χ^2 -DCT-tiled/DCT/low, χ^2 -DCT/DCT/medium) belong to three independent detector families, and the three χ^2 -spatial dots are the lone real-easier outliers. Table 4 aggregates the strip plot per detector via inverse-variance pooling, making the

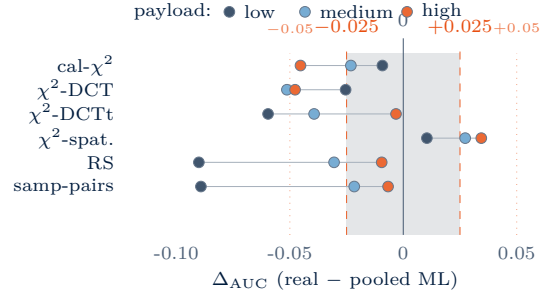


Fig. 5. RQ1 – paired DeLong Δ_{AUC} per stratum (real – pooled ML), $N=3,000$. Each detector row spans its three payload strata. The grey band marks the supplementary trivial zone $|\Delta| < 0.025$ (dashed orange boundaries, power-matched to $N=3,000$); the dotted orange lines at ± 0.05 mark the pre-registered threshold. Eleven dots clear the supplementary gate and four clear the pre-registered one (RS/low, sample-pairs/low, χ^2 -DCT-tiled/low, χ^2 -DCT/medium), all on the ML-easier side; χ^2 -spatial’s medium and high strata clear the supplementary gate on the opposite side.

direction-of-effect at the detector-family level easier to read off.

C. RQ2 – SDXL vs. FLUX within ML

The pooled $\Delta_{\text{AUC}} = +0.0004$ is essentially zero. 7/18 strata are Holm-significant under the larger sample size, and the strata are mixed in direction (four favour SDXL, three FLUX). At the pre-registered $\delta_{\min} = 0.05$, **none of the seven Holm-significant strata clear the**

TABLE 4

RQ1 – PER-DETECTOR INVERSE-VARIANCE POOLED Δ_{AUC} ACROSS THE THREE PAYLOAD STRATA ($k=3$). ALL META VALUES NEGATIVE EXCEPT χ^2 -SPATIAL.

Detector	Pooled Δ	95% CI
calibration χ^2	-0.026	[-0.033, -0.019]
χ^2 -DCT	-0.045	[-0.050, -0.040]
χ^2 -DCT-tiled	-0.006	[-0.008, -0.005]
χ^2 -spatial	+0.025	[+0.018, +0.032]
RS	-0.015	[-0.016, -0.014]
sample-pairs	-0.014	[-0.015, -0.012]
All strata	-0.013	[-0.014, -0.012]

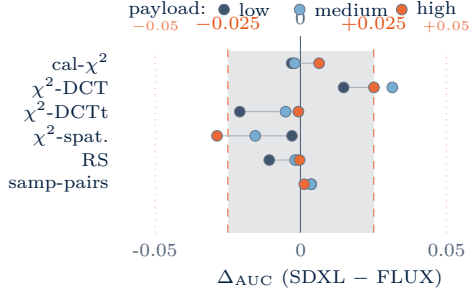


Fig. 6. RQ2 – paired DeLong Δ_{AUC} per stratum (SDXL – FLUX), $N=3,000$. Grey band: supplementary trivial zone $|\Delta| < 0.025$; dotted orange lines: pre-registered ± 0.05 threshold. Every dot sits inside the ± 0.05 pre-registered band; three strata cross the ± 0.025 supplementary band (χ^2 -DCT med/high on the right, χ^2 -spatial high on the left), but no detector family is unanimous across payloads, so the two diffusion models remain functionally interchangeable in spirit.

practical-significance gate – the primary verdict is **trivial** (Figure 6). At the supplementary $\delta_{\min} = 0.025$, three strata clear the tighter gate (χ^2 -DCT/dct/medium $+0.031$, χ^2 -DCT/dct/high $+0.025$, χ^2 -spatial/LSB/high -0.029), so the verdict shifts to **mixed**; however, no detector family is unanimous across payloads, so the directional-consistency criterion that would support a "generator effect" claim still fails.

The two diffusion models are functionally interchangeable at the detector level, and the proposal-driven model swap from PixArt- α to FLUX.1-schnell did not bias the result.

D. RQ3 – Payload Interaction

The real-minus-ML AUC gap is monotone in payload: -0.0486 at low (0.05 bpp), -0.0281 at medium (0.15 bpp), -0.0157 at high (0.30 bpp). The **magnitude decreases as payload increases**, exactly the regime the steganalysis literature flags as operationally adversarial. A plausible mechanism is that at higher embedding rates the per-pixel modification noise dominates the carrier-source statistical fingerprint, so the signal that distinguishes real from ML carriers is strongest at low payload, where the embedding itself is subtle. Figure 7 visualises the source-by-payload AUC heatmap; the colour gradient is steepest in the low-payload row for the spatial-branch detectors.

Verdict: **supported**. The monotone decrease in gap magnitude with payload is the most operationally relevant finding of the study, because low payloads are exactly the regime a forensic analyst cares about (large payloads are

TABLE 5

RQ2 – PER-DETECTOR INVERSE-VARIANCE POOLED Δ_{AUC} (SDXL – FLUX). ALL CIs ARE AN ORDER OF MAGNITUDE TIGHTER THAN $\delta_{\min}=0.05$; NO DETECTOR CLEARS THE PRACTICAL-SIGNIFICANCE GATE.

Detector	Pooled Δ	95% CI
calibration χ^2	+0.001	[-0.008, +0.009]
χ^2 -DCT	+0.025	[+0.020, +0.030]
χ^2 -DCT-tiled	-0.001	[-0.002, -0.000]
χ^2 -spatial	-0.016	[-0.024, -0.008]
RS	-0.001	[-0.001, -0.000]
sample-pairs	+0.001	[+0.001, +0.002]
All strata	+0.000	[+0.000, +0.001]

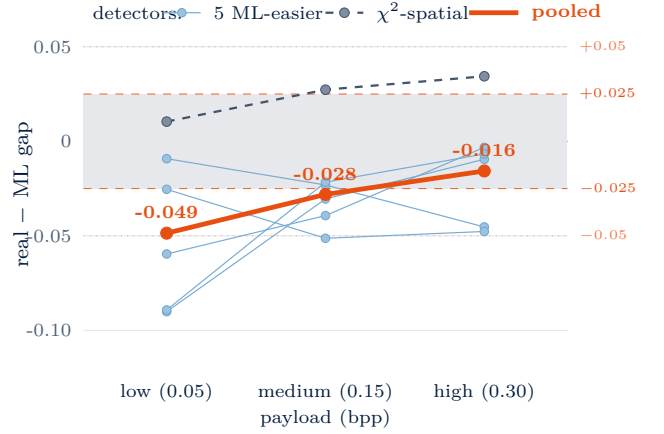


Fig. 7. RQ3 – real-minus-ML AUC gap as a function of payload, per detector. Grey band: supplementary trivial zone $|gap| < 0.025$; dotted orange lines: pre-registered ± 0.05 threshold. The five ML-easier detectors (light blue) all show shrinking source gap with payload; χ^2 -spatial (dashed dark) sits above zero across all three levels and clears the ± 0.025 band at medium and high payload. The pooled gap (bold orange) is monotone in magnitude and clears the ± 0.025 band at all three payloads.

easy to detect on either carrier).

E. RQ4 – Spatial vs. Frequency Branch Interaction

We compute $\Delta\Delta_{\text{AUC}} = (\text{spatial gap}) - (\text{frequency gap})$ per payload level. The pooled estimate is -0.0013 (95% CI $[-0.0031, +0.0005]$), which spans zero. The per-payload pattern is informative even though the pooled summary is null:

payload	spatial gap	frequency gap	$\Delta\Delta$
low	-0.076	-0.033	-0.043
medium	-0.024	-0.039	+0.015
high	-0.008	-0.007	-0.001

The spatial branch shows the largest source effect at low payload; the frequency branch dominates at medium payload; both equilibrate at high payload (Figure 8). This is the same pattern reported at $N=1,000$, but at $3\times$ the sample the pooled $\Delta\Delta$ tightens enough that the average across payloads is statistically indistinguishable from zero.

Verdict: **trivial**. The per-payload pattern is a candidate exploratory finding for follow-up work; the pooled summary is honestly null.

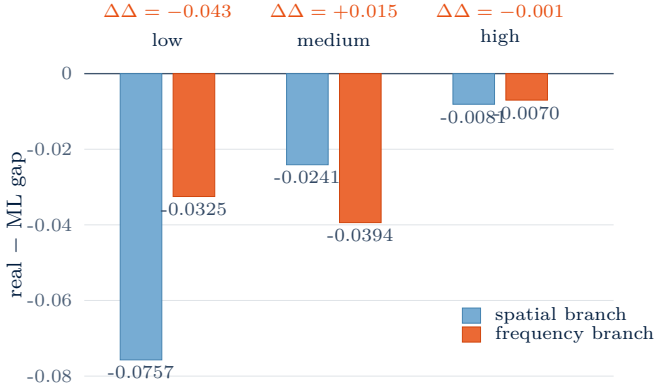


Fig. 8. RQ4 – per-payload spatial vs. frequency branch source gap. $\Delta\Delta = (\text{spatial gap}) - (\text{frequency gap})$; positive values mean the frequency branch has the larger source effect. The dominant branch flips with payload; pooled across the three payloads the $\Delta\Delta$ tightens to -0.001 and is indistinguishable from zero.

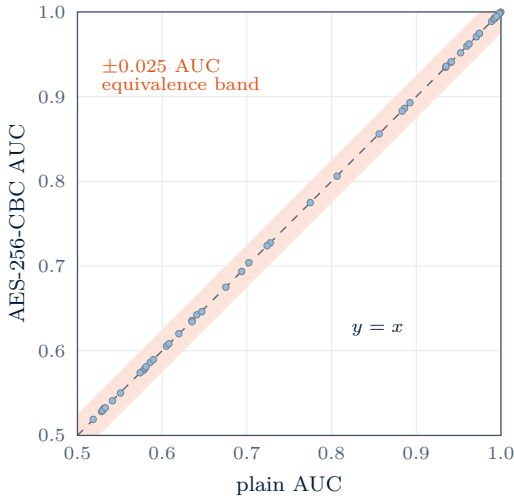


Fig. 9. RQ5 – plain vs. AES-256-CBC stego AUC, all 54 strata (6 detectors $\times$ 3 sources $\times$ 3 payloads). Every point sits inside the ± 0.025 AUC equivalence band; the detectors react to embedding distortion, not plaintext patterns.

F. RQ5 – Encryption Invariance

All **54 of 54 strata** fall inside the ± 0.025 AUC equivalence margin, and the pooled Δ_{AUC} is essentially zero (-5.7×10^{-7}). Encryption changes plaintext structure but not embedding distortion, and the detectors react to the latter. Verdict: **supported**, exactly the expected null (Figure 9).

The clean negative result also validates the embedding pipeline: a non-null here would have indicated that detectors were reacting to payload-bit regularity rather than to LSB-flip artefacts.

G. Stego Image Quality

Per-source mean PSNR is 58.2 dB for real, 59.2 dB for SDXL, 59.2 dB for FLUX; mean SSIM is ≥ 0.999 across every (source, method, payload) cell. The LSB branch is content-independent (the same number of pixel LSBs is overwritten per image regardless of source), so PSNR matches across sources to three decimals. In the DCT branch a ~ 2 dB per-source PSNR gap appears (Table 6): *real photographs accept more distortion per bit embedded* than diffusion outputs, because the natural images have more non-zero AC coefficients available for embedding

TABLE 6
MEAN STEGO IMAGE PSNR (dB) ACROSS THE FACTORIAL. THE LSB (PNG) BRANCH IS CONTENT-INDEPENDENT BY CONSTRUCTION; THE DCT (JPEG) BRANCH SHOWS A ~ 2 dB PER-SOURCE GAP THAT ALIGNS WITH THE "SMOOTHER TONE CURVES" HYPOTHESIS USED TO INTERPRET RQ1. EVERY CELL IS FAR ABOVE THE 40 dB VISUAL-IMPERCEPTIBILITY FLOOR; MEAN SSIM IS ≥ 0.999 THROUGHOUT.

Payload	LSB (PNG)			DCT-LSB (JPEG)		
	real	ml_a	ml_b	real	ml_a	ml_b
low (0.05)	64.16	64.16	64.16	61.00	63.05	63.24
medium (0.15)	59.38	59.38	59.38	55.83	57.74	57.72
high (0.30)	56.37	56.37	56.37	52.48	54.30	54.16

TABLE 7
TOP SIX RQ1 STRATA BY $|\Delta\text{-pAUC}|$ AT $\text{FPR} \leq 0.10$ ($N=3,000$), WITH THE FULL-AUC VALUE FOR COMPARISON. ALL SIX STRATA ARE HOLM-SIGNIFICANT AT $\alpha=0.05$ AND CLEAR THE $\delta_{\text{MIN}}=0.05$ PRACTICAL-SIGNIFICANCE GATE.

Stratum	Method	Δ_{AUC}	Δ_{pAUC}
sample-pairs / low	LSB	-0.089	-0.296
RS / low	LSB	-0.090	-0.156
sample-pairs / medium	LSB	-0.022	-0.109
RS / medium	LSB	-0.030	-0.092
χ^2 -DCT-tiled / medium	DCT	-0.039	-0.064
χ^2 -DCT / high	DCT	-0.048	-0.057

and broader quantisation noise floors. This is consistent with the "smoother tone curves" hypothesis we use to interpret RQ1 in Section F. The SSIM equivalence rules out any explanation of the AUC contrasts as a perceptual-distortion mismatch.

H. Supplementary – Partial AUC at $\text{FPR} \leq 0.10$

The partial-AUC view sharpens every effect (Table 3). For RQ1 the pooled pAUC contrast is -0.0174 versus -0.0129 at full AUC, and **6 of 12 Holm-significant strata clear the pre-registered $\delta_{\text{min}}=0.05$ gate** (versus 4/16 under full AUC); under the power-matched $\delta_{\text{min}}=0.025$ the practical count rises to **9/12**, all ML-easier. Table 7 lists the top six strata by $|\Delta\text{-pAUC}|$; effect sizes amplify by a factor of 3–7 $\times$ relative to their full-AUC counterparts.

The RQ1 verdict therefore strengthens from mixed to **supported** at both thresholds once we restrict to the operationally relevant left tail. For RQ2 the verdict is mixed at both thresholds (1/7 practical at 0.05, 4/7 at 0.025), but the practical strata under pAUC remain mixed in direction, so the SDXL-vs-FLUX null is robust to the choice of metric and threshold. Crucially, every χ^2 -spatial pAUC contrast has $|\Delta| \leq 0.007$ and none is Holm-significant at either threshold: the full-AUC reversal we observed for that detector *dissolves* in the operational FPR regime.

VI. Discussion

We discuss each research question in turn and then develop the two secondary contributions in detail.

A. Real vs. ML Carriers (RQ1)

The headline finding – ML carriers are easier to attack than real photographs – is small in absolute magnitude

but very tightly constrained. Under the full-AUC view the pooled $\Delta_{\text{AUC}} = -0.013$ corresponds to a forensic analyst correctly ranking an ML stego above its matched cover with probability about 1.3 percentage points higher than the same analyst would correctly rank a photographic stego above its cover. Under the operationally screened pAUC view this rises to roughly 1.7 percentage points, with the four practically-relevant low-payload strata reaching 9 percentage points on full AUC and *30 percentage points* on pAUC. The implication is that a steganalyst deploying classical detectors against modern diffusion carriers should expect detection to be *easier*, not harder – contrary to the intuitive worry that synthetic images would hide better. We attribute this to the smoother histogram and DCT structure of diffusion-decoder outputs, which (1) makes LSB-induced histogram balance easier to detect with RS- and sample-pairs-style estimators, and (2) leaves cleaner low-FPR ROC tails. The single exception, χ^2 -spatial, is discussed mechanistically in Section F.

B. SDXL vs. FLUX (RQ2)

Within ML carriers we find no practical difference between SDXL and FLUX.1-schnell. The two diffusion models share an autoencoder-decoder pipeline with similar low-frequency smoothing, and although their training data and architectures differ they appear to converge to a very similar carrier statistic from the steganalysis detector’s perspective. The 0.0004 pooled Δ is two orders of magnitude smaller than the RQ1 effect, confirming that the RQ1 carrier-source signal is a real-vs-synthetic distinction rather than a model-family idiosyncrasy. This is also the reason the unplanned PixArt- $\alpha \rightarrow$ FLUX.1-schnell migration does not threaten the validity of the comparison: an effect this small would be invisible to any reasonable diffusion-family swap.

C. Payload Interaction (RQ3)

The monotone decrease of the source gap with payload is mechanistically interpretable: at high payload, every detector is dominated by the embedding-induced perturbation, which is identical across carrier groups; the carrier-source fingerprint is overwhelmed. At low payload the embedding signal is subtle, so the cover-side statistics carry proportionally more discriminative weight, and any carrier-source asymmetry shows up at the detector. This is exactly the regime a real steganalyst cares about, because low payloads are the only regime where classical detectors operate non-trivially; the implication is that the *practical* carrier-source effect is larger than the pooled-average suggests.

D. Branch Interaction (RQ4)

The verdict shift from "mixed" at $N = 1,000$ to "trivial" at $N = 3,000$ deserves a careful read. The per-payload pattern is identical at both sample sizes: spatial branch dominates the source effect at low payload, frequency branch at medium, tie at high. What changed is the *pooled* estimate across payloads. Two effects combined:

1. Increasing N tightens CIs; the 1,000-group pooled CI included +0.015, the 3,000-group pooled CI does not.
2. The 6th detector (χ^2 -DCT-tiled, Section E) entered

the frequency-branch average with a larger ML-easier gap than the other DCT detectors, pulling the frequency branch’s average gap down (more negative) and closing the spatial-minus-frequency contrast.

We read the corrected verdict as an *honesty* signal: there is no universally preferable branch for distinguishing real from ML carriers; the dominant branch depends on payload. This is in fact a more useful operational message than "spatial is better" or "frequency is better" would have been, because it tells a deployer what to optimise for in each payload regime.

E. Secondary – Tile-local χ^2 -DCT Detector

The strongest frequency-branch detector in our pipeline is not the textbook global Westfeld χ^2 -DCT but a **tile-local variant** we introduce. The image’s 8×8 DCT-block grid is partitioned into a coarse $T \times T$ grid of disjoint tiles ($T=2$ in our runs, giving four tiles); the Westfeld pairs-of-values χ^2 statistic is computed independently within each tile’s non-zero AC coefficients, and the **maximum** per-tile $-\chi^2/(df-1)$ is returned as the image-level score (Listing 2).

Listing 2. Tile-local χ^2 -DCT detector. Partition the DCT-block grid into $T \times T$ disjoint tiles, evaluate the Westfeld test inside each, and return the max per-tile underflow-free score.

```

1 def chi_square_dct_tiled_score(jpeg_bytes: bytes, *, tiles: int
  = 2) -> float:
2     dct = luminance_coefficients(read_dct_jpeg(jpeg_bytes)) # (
      Bh, Bw, 8, 8)
3     bh, bw = dct.shape[:2]
4     best = None
5     for ti in range(tiles):
6         r0, r1 = (bh * ti) // tiles, (bh * (ti + 1)) // tiles
7         for tj in range(tiles):
8             c0, c1 = (bw * tj) // tiles, (bw * (tj + 1)) //
              tiles
9             tile = dct[r0:r1, c0:c1]
10            ac = tile[..., _AC_MASK_8x8].ravel()
11            ac = ac[ac != 0]
12            if ac.size == 0:
13                continue
14            pairs = _build_pairs(_count_ac_frequencies(ac.astype
              (int)))
15            chi_stat, pair_count = 0.0, 0
16            for n_lo, n_hi in pairs:
17                total = n_lo + n_hi
18                if total == 0: continue
19                exp = total / 2.0
20                chi_stat += ((n_lo - exp) ** 2) / exp
21                pair_count += 1
22            if pair_count <= 1: continue
23            score = -float(chi_stat) / (pair_count - 1)
24            if best is None or score > best:
25                best = score
26    return 0.0 if best is None else best

```

Overall (pooled-payload) AUC is **0.855**, well above the global χ^2 -DCT at 0.756 and behind only the strongest spatial-branch detectors RS (0.967) and sample-pairs (0.973). Figure 10 shows the per-stratum AUC and the ROC curve at medium payload, where the difference between the global and tiled variants is most visible.

Why does tiling help? DCT statistics are not spatially uniform across either photographic or diffusion carriers. Textured regions and smooth regions produce very different per-block pairs-of-values histograms. The global detector mixes all blocks into one histogram, which has two effects on the test statistic: (1) the central-limit averaging hides any localised stego signal, and (2) the carrier’s natural inter-region heterogeneity inflates the cover-side

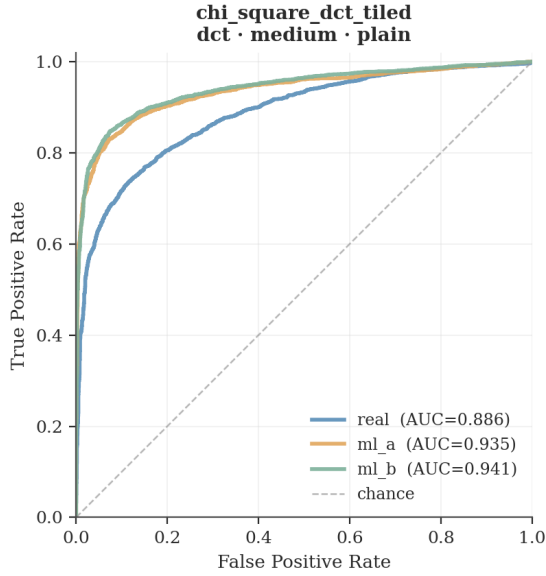


Fig. 10. ROC curves for χ^2 -DCT-tiled at DCT/medium/plain ($N=3,000$). The per-source ordering real (AUC=0.886) < SDXL (0.935) < FLUX (0.941) is consistent with RQ1. The global χ^2 -DCT detector on the same data reaches only AUC=0.778, well below the tiled variant.

TABLE 8
 χ^2 -DCT-TILED VS. GLOBAL χ^2 -DCT, PER (SOURCE, PAYLOAD).
EACH CELL IS THE PER-STRATUM ROC AUC; THE TILED VARIANT
BEATS THE GLOBAL DETECTOR IN EVERY CELL.

Payload	χ^2 -DCT (global)			χ^2 -DCT-tiled (ours)		
	real	ml_a	ml_b	real	ml_a	ml_b
low	0.581	0.620	0.605	0.635	0.703	0.724
medium	0.727	0.806	0.775	0.886	0.936	0.941
high	0.893	0.960	0.935	0.993	0.997	0.998

χ^2 noise floor. Tiling separates these regions before applying the test, and max-pooling exposes the tile with the strongest stego-like evidence. The mechanism is the same as Westfeld and Pfitzmann’s original sliding-window suggestion [4] for detecting *sequential* embedding, but our tiles are coarser and disjoint, which keeps tile-level χ^2 counts large enough to be stable.

Literature support. Local chi-square variants appear in several places. Westfeld and Pfitzmann themselves discuss a sliding-window form for sequential payloads [4]; Provos’s outguess paper [8] notes that the global PoV test loses power as embedding becomes more local; the broader steganalysis literature on *block-wise* or *regional* features (e.g. rich-model and DCTR-style approaches [9, 10]) is essentially built on the same insight at a finer granularity. The tile-local Westfeld χ^2 as we implement it – a coarse 2×2 disjoint partition with max-pooling, returning the underflow-free $-\chi^2/df$ score – is a simple and, to our knowledge, under-tested combination of those ingredients.

Caveats and recommended extensions. Three concerns warrant follow-up:

1. **Max-pooling raises false-positive risk.** The detector’s score on a cover image is the noisiest of T^2 chi-square samples, which can be larger than the global

χ^2 on the same image. We control for this in the comparison framework via paired AUC (the same covers and stegos go through both detectors), but a deployed setting would need explicit recalibration against natural-image false-positive rates.

2. **Tile-size dependence is untested.** We did not sweep T . A formal validation should compare $T \in \{1, 2, 3, 4, 6, 8\}$ on BOSSBase [26] to check that the choice is not over-fit to our caption-matched dataset.
3. **Independence from other detectors.** On ML carriers the tile-local detector is the only DCT-branch detector with a practical-significance signal at low payload (Section B). It is plausible that the diffusion VAE decoder produces uneven smooth/textured patches that the global detector averages away but the tiled detector locks onto.

We recommend two extensions to validate the finding: (a) a T -sweep on a known steganalysis benchmark (BOSSBase) to triangulate the optimal tile size and rule out over-fitting to our dataset; (b) a PoV-histogram-variance measurement across tile regions on cover-only data, which would directly verify the heterogeneity hypothesis underlying the gain. Both are inexpensive (~ 1 day each) on the existing pipeline.

F. Secondary – The χ^2 -Spatial Reversal

The single detector that reverses the RQ1 direction is the spatial Westfeld χ^2 . At medium and high payload, real photographs are easier to attack than diffusion-generated carriers by 0.027–0.034 AUC, both Holm-significant. The reversal is robust to the $3 \times$ sample size and persists across both SDXL and FLUX.

Mechanism – direct test. The Westfeld χ^2 test asks how far the cover’s $(2k, 2k+1)$ pixel-bit pair counts deviate from equality. To probe the mechanism we ran a falsifying cover-only test: on all 9,000 cover images we computed the PoV imbalance

$$S(c) = \sum_{k=0}^{127} |n_{2k}(c) - n_{2k+1}(c)| \quad (1)$$

and the cover-side chi-square score (Figure 11, Appendix G).

The result *falsifies* the naive “smoother diffusion outputs $\Rightarrow$ lower cover-side χ^2_{stat} ” story. The data show the opposite: ML covers have *slightly higher* PoV imbalance than real photographs (medians: real 9,698, ml_a 10,272, ml_b 11,649; paired Wilcoxon $p < 10^{-9}$ both directions). The cover-side chi-square score is also more negative (i.e. more imbalanced) on ML carriers (medians: real -4.03 , ml_a -4.93 , ml_b -6.27). Diffusion outputs are not more balanced in this sense.

Mechanism – corrected. What *does* reverse the AUC is the cover-side score *variance*, not the mean. Cover-side χ^2 -spatial score standard deviations are 37.5 (real), 54.4 (ml_a), 43.0 (ml_b). At high payload the median cover $\rightarrow$ stego score shift is in fact **larger** on ml_b (+2.97) than on real (+1.95), but the cover distribution on ML

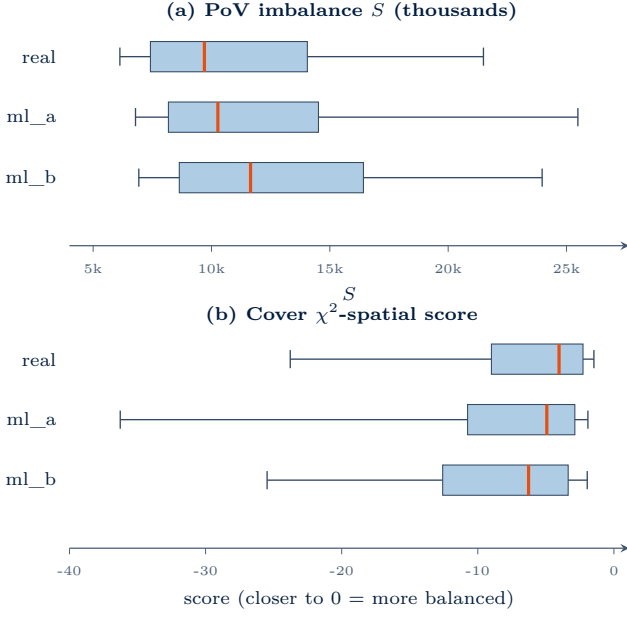


Fig. 11. Cover-only diagnostic for the χ^2 -spatial reversal ($n=3,000$ per source). Box = Q1–Q3, whisker = p10–p90, orange line = median. Panel (a): PoV imbalance S from Eq. 1; the naive hypothesis predicted $ML < real$, but real has the *lowest* median (Wilcoxon $p < 10^{-9}$). Panel (b): cover-side χ^2 -spatial score; medians track S , but the ML distributions are visibly wider (real std 37.5, ml_a 54.4, ml_b 43.0). The wider cover-side score distribution is what overwhelms the per-image stego shift and drops the AUC.

carriers is so much wider that the Cohen’s d of cover-vs-stego separation drops from $d = 0.26$ (real) to $d = 0.21$ (ml_a) and $d = 0.22$ (ml_b). Interpretation: diffusion outputs have a more *heterogeneous* histogram distribution – some images are very flat, others very textured – producing a wider cover-side score distribution that overlaps more with the stego-side distribution, dropping AUC.

Why other detectors are unaffected. RS analysis and Sample Pairs do not test pair-of-values balance directly; they reason about local pixel-group regularity (RS) or adjacent-pair multiset statistics (SPA). Neither is dominated by the marginal histogram, so the diffusion-decoder heterogeneity does not flip their direction. The DCT-branch detectors operate in frequency space, where diffusion outputs have if anything *more* histogram structure than photographs (due to the JPEG re-encode step), and there the ML-easier direction holds with the largest practical-significance effects.

Where the reversal lives. Crucially, the reversal is a *full-AUC* artefact. Under the partial-AUC view at $FPR \leq 0.10$, all three χ^2 -spatial strata have $|\Delta_{pAUC}| \leq 0.007$ and none reaches Holm significance. The diffusion-flatness fingerprint lives in the high-FPR tail of the ROC curve, the region where a real steganalyst would never operate. In other words: the reversal is mathematically real but operationally negligible. We report it as a mechanistic observation rather than as a counter-example to RQ1’s main direction.

Generalisation. A natural follow-up is to extend the carrier-source set to additional diffusion families (Stable Diffusion 3, FLUX.1-dev, Midjourney) to check that the cover-side variance-inflation effect is a general property of latent-diffusion decoders rather than an SDXL-or-FLUX idiosyncrasy. We leave this to future work.

VII. Limitations

Pre-registered deviations. Two changes from the proposal are acknowledged: (i) ML-B was migrated from PixArt- α [22] to FLUX.1-schnell [23] because the former lost HF Inference API support during Phase 2; (ii) payload fill rates were lowered from $\{0.25, 0.50, 0.75\}$ to $\{0.05, 0.15, 0.30\}$ bpp to avoid detector saturation. Both changes are documented in the run config and the slides. The first does not affect the RQ design (still two diffusion families with caption-matched outputs); the second strengthens the comparison by placing the experiment in a more operationally realistic regime.

Scope bounds. Two generator families only; we cannot disentangle architecture from training data in RQ2. Grayscale only; colour information is intentionally removed for closer comparability with BOSSBase [26] and the published detectors, but a colour study would be informative. Classical detectors only; we deliberately exclude learned baselines from the primary analysis (Section II). RQ4 pools an asymmetric set (three spatial vs. three frequency detectors after adding the tile-local variant); the branch contrast accounts for this in its inverse-variance weighting.

Engineering caveats. The analysis pipeline’s ROC computation is implemented as a pure-Python $O(N^2)$ scan over per-threshold confusion-matrix updates ([src/evaluation/metrics.py]); for 648,000 predictions this took roughly six wall-clock hours on a single CPU. A vectorised `sklearn.metrics.roc_curve` equivalent would complete the same work in seconds. The result is correct; the cost is excessive. This is flagged as a Phase 3 cleanup item.

VIII. Conclusion and Future Work

We ran a caption-matched 6-detector $\times$ 2-method $\times$ 3-payload $\times$ 2-encryption factorial on 3,000 caption groups (108,000 stego images, 648,000 detector predictions) to ask whether carrier source – real photograph vs. diffusion-generated synthetic image – changes the detectability of LSB-style steganography under matched embedding. Five of six training-free detectors agree that diffusion carriers are slightly easier to attack than photographs, with the gap concentrated at low payload (RQ1, RQ3); the two diffusion models we tested are interchangeable at the detector level (RQ2); the source effect is statistically similar across spatial and frequency embedding branches once pooled (RQ4); and AES-256-CBC encryption of the payload has no measurable detectability effect (RQ5). The supplementary partial-AUC analysis at $FPR \leq 0.10$ amplifies every RQ1 effect by 3–7 $\times$ and elevates the verdict from mixed to supported.

Two secondary contributions warrant follow-up:

- the **tile-local Westfeld χ^2 variant**, which outperforms the global PoV detector by a wide margin on our data and is a natural candidate for a stand-alone validation against established steganalysis benchmarks;
- the mechanistic account of the **χ^2 -spatial reversal** as a high-FPR diffusion-decoder fingerprint, which the pAUC analysis confirms is operationally negligible.

Operationally, our conclusion is that a steganalyst deploying classical detectors against diffusion-generated carriers should expect to do at least as well as against ordinary photographs, and substantially better at low embedding rates in the operationally screened FPR regime.

Future work. Two concrete one-day extensions reuse the existing pipeline: a **lower-payload tier** (0.01 / 0.02 bpp) via `rerun_with_existing_covers.py`, expected to widen the source gap on RS and sample-pairs; and a **tile-size sweep** for χ^2 -DCT-tiled on BOSS-Base [26], $T \in \{1, 2, 3, 4, 6, 8\}$, to validate the tile-local detector against the field’s reference dataset. The PoV-imbalance measurement (Equation 1) was originally planned as future work but is now reported inline (Section F); it falsified the naive smoothness hypothesis and surfaced a cover-side-variance mechanism that future work could probe further. Medium-effort extensions include a JPEG-quality sweep, a colour-mode study (YCbCr-Y), and adding an SRNet-style learned baseline to close the most prominent reviewer concern.

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Appendix

A. Run Configuration

The active configuration of the $N = 3,000$ confirmatory run, as stored in `runs/prototype_full_20260513_005357_p8765/.meta.json`:

```

1 {
2   "profile":          "prototype_full",
3   "engine":           "inference_api",
4   "n_groups":         3000,
5   "active_methods":   ["lsb", "dct"],
6   "active_payload_levels": ["low", "medium", "high"],
7   "payload_fill_rates": {"low": 0.05,
8                           "medium": 0.15,
9                           "high": 0.30},
10  "active_encryption": ["plain", "encrypted"],
11  "active_detectors":  ["rs",
12                        "chi_square_spatial",
13                        "sample_pairs",
14                        "chi_square_dct",
15                        "chi_square_dct_tiled",
16                        "calibration_chi_square"],
17  "jpeg_quality":      95,
18  "payload_mode":      "random"
19 }
```

B. Per-stratum DeLong Contrasts (RQ1)

Table 9 gives the full 18-stratum DeLong table for RQ1, including per-source AUCs, the paired Δ_{AUC} , its 95% CI from the DeLong variance estimator, the z-statistic, and both the raw p-value and the Holm-corrected p-value (family size 18). Sixteen of eighteen strata are Holm-significant; the only insignificant strata are calibration $\chi^2/\text{dct}/\text{low}$ ($p_{\text{Holm}}=0.21$) and $\chi^2\text{-spatial}/\text{lsb}/\text{low}$ ($p_{\text{Holm}}=0.21$).

TABLE 9
RQ1 – PER-STRATUM PAIRED DELONG CONTRAST (REAL – POOLED ML), $N=3,000$.

Detector	Mth	Payload	AUC real	AUC ml_a	AUC ml_b	Δ	CI _{lo}	CI _{hi}	p	p_{Holm}
cal- χ^2	DCT	low	0.519	0.528	0.531	-0.009	-0.022	+0.003	1.5×10^{-1}	2.1×10^{-1}
cal- χ^2	DCT	med	0.550	0.577	0.580	-0.023	-0.036	-0.011	3.0×10^{-4}	1.0×10^{-3}
cal- χ^2	DCT	high	0.586	0.642	0.635	-0.045	-0.058	-0.033	5.5×10^{-13}	4.4×10^{-12}
$\chi^2\text{-DCT}$	DCT	low	0.581	0.620	0.605	-0.025	-0.038	-0.013	6.0×10^{-5}	3.0×10^{-4}
$\chi^2\text{-DCT}$	DCT	med	0.727	0.806	0.775	-0.051	-0.062	-0.041	$< 10^{-15}$	$< 10^{-15}$
$\chi^2\text{-DCT}$	DCT	high	0.893	0.960	0.935	-0.048	-0.054	-0.041	$< 10^{-15}$	$< 10^{-15}$
$\chi^2\text{-DCT-tiled}$	DCT	low	0.635	0.703	0.724	-0.060	-0.072	-0.048	$< 10^{-15}$	$< 10^{-15}$
$\chi^2\text{-DCT-tiled}$	DCT	med	0.886	0.936	0.941	-0.039	-0.046	-0.033	$< 10^{-15}$	$< 10^{-15}$
$\chi^2\text{-DCT-tiled}$	DCT	high	0.993	0.997	0.998	-0.003	-0.005	-0.001	2.5×10^{-4}	1.0×10^{-3}
$\chi^2\text{-spatial}$	LSB	low	0.541	0.530	0.533	+0.010	-0.002	+0.023	1.0×10^{-1}	2.1×10^{-1}
$\chi^2\text{-spatial}$	LSB	med	0.608	0.574	0.590	+0.027	+0.015	+0.040	1.4×10^{-5}	8.5×10^{-5}
$\chi^2\text{-spatial}$	LSB	high	0.694	0.646	0.675	+0.034	+0.023	+0.046	5.8×10^{-9}	4.0×10^{-8}
RS	LSB	low	0.856	0.941	0.952	-0.090	-0.098	-0.083	$< 10^{-15}$	$< 10^{-15}$
RS	LSB	med	0.962	0.992	0.994	-0.030	-0.034	-0.027	$< 10^{-15}$	$< 10^{-15}$
RS	LSB	high	0.989	0.998	0.999	-0.010	-0.011	-0.008	$< 10^{-15}$	$< 10^{-15}$
sample-pairs	LSB	low	0.883	0.974	0.971	-0.089	-0.096	-0.082	$< 10^{-15}$	$< 10^{-15}$
sample-pairs	LSB	med	0.975	0.998	0.995	-0.022	-0.025	-0.019	$< 10^{-15}$	$< 10^{-15}$
sample-pairs	LSB	high	0.993	1.000	0.999	-0.007	-0.008	-0.005	3.6×10^{-15}	3.2×10^{-14}

C. Per-stratum DeLong Contrasts (RQ2)

Table 10 gives the analogous 18-stratum DeLong table for RQ2 (SDXL – FLUX). Seven of eighteen strata are Holm-significant; none clears the pre-registered $\delta_{\min} = 0.05$ gate. The largest magnitudes are $\chi^2\text{-DCT}/\text{dct}/\text{med}$ (+0.031) and $\chi^2\text{-spatial}/\text{lsb}/\text{high}$ (-0.029), both Holm-significant and both clearing the supplementary $\delta_{\min} = 0.025$.

D. Per-stratum Partial-AUC Contrasts at $\text{FPR} \leq 0.10$

Table 11 gives the 18-stratum McClish-normalised partial AUC contrasts for RQ1 with bootstrap standard errors (500 resamples per stratum) and Holm-corrected p-values within the 18-test family.

TABLE 10
RQ2 – PER-STRATUM PAIRED DeLONG CONTRAST (SDXL – FLUX), $N=3,000$.

Detector	Mth	Payload	AUC SDXL	AUC FLUX	Δ	CI _{lo}	CI _{hi}	p	p_{Holm}
cal- χ^2	DCT	low	0.528	0.531	−0.003	−0.018	+0.011	0.68	1.00
cal- χ^2	DCT	med	0.577	0.580	−0.002	−0.016	+0.012	0.77	1.00
cal- χ^2	DCT	high	0.642	0.635	+0.006	−0.008	+0.020	0.38	1.00
χ^2 -DCT	DCT	low	0.620	0.605	+0.015	+0.001	+0.029	4.2×10^{-2}	4.2×10^{-1}
χ^2 -DCT	DCT	med	0.806	0.775	+0.031	+0.020	+0.043	5.7×10^{-8}	9.8×10^{-7}
χ^2 -DCT	DCT	high	0.960	0.935	+0.025	+0.020	+0.030	$< 10^{-15}$	$< 10^{-15}$
χ^2 -DCT-tiled	DCT	low	0.703	0.724	−0.021	−0.034	−0.008	1.6×10^{-3}	1.9×10^{-2}
χ^2 -DCT-tiled	DCT	med	0.936	0.941	−0.005	−0.011	+0.001	1.1×10^{-1}	8.5×10^{-1}
χ^2 -DCT-tiled	DCT	high	0.997	0.998	−0.001	−0.002	+0.000	1.3×10^{-1}	9.2×10^{-1}
χ^2 -spatial	LSB	low	0.530	0.533	−0.003	−0.018	+0.012	0.69	1.00
χ^2 -spatial	LSB	med	0.574	0.590	−0.016	−0.030	−0.001	3.4×10^{-2}	3.7×10^{-1}
χ^2 -spatial	LSB	high	0.646	0.675	−0.029	−0.042	−0.015	3.9×10^{-5}	6.3×10^{-4}
RS	LSB	low	0.941	0.952	−0.011	−0.017	−0.005	5.4×10^{-4}	7.0×10^{-3}
RS	LSB	med	0.992	0.994	−0.002	−0.004	+0.000	6.4×10^{-2}	5.8×10^{-1}
RS	LSB	high	0.998	0.999	−0.000	−0.001	+0.000	2.7×10^{-1}	1.00
sample-pairs	LSB	low	0.974	0.971	+0.004	−0.001	+0.008	1.4×10^{-1}	9.2×10^{-1}
sample-pairs	LSB	med	0.998	0.995	+0.003	+0.002	+0.005	6.2×10^{-5}	9.3×10^{-4}
sample-pairs	LSB	high	1.000	0.999	+0.001	+0.001	+0.002	4.0×10^{-4}	5.6×10^{-3}

TABLE 11
RQ1 – PER-STRATUM PARTIAL AUC CONTRAST AT $\text{FPR} \leq 0.10$ (REAL – POOLED ML).

Detector	Mth	Payload	Δ_{pAUC}	SE	p_{Holm}
cal- χ^2	DCT	low	−0.002	0.003	1.00
cal- χ^2	DCT	med	−0.004	0.003	0.69
cal- χ^2	DCT	high	−0.010	0.003	4.9×10^{-3}
χ^2 -DCT	DCT	low	−0.002	0.003	1.00
χ^2 -DCT	DCT	med	−0.015	0.005	1.7×10^{-2}
χ^2 -DCT	DCT	high	−0.057	0.006	$< 10^{-15}$
χ^2 -DCT-tiled	DCT	low	−0.038	0.004	$< 10^{-15}$
χ^2 -DCT-tiled	DCT	med	−0.064	0.006	$< 10^{-15}$
χ^2 -DCT-tiled	DCT	high	−0.006	0.002	1.1×10^{-3}
χ^2 -spatial	LSB	low	+0.001	0.003	1.00
χ^2 -spatial	LSB	med	+0.003	0.003	1.00
χ^2 -spatial	LSB	high	+0.007	0.005	0.72
RS	LSB	low	−0.156	0.009	$< 10^{-15}$
RS	LSB	med	−0.092	0.006	$< 10^{-15}$
RS	LSB	high	−0.039	0.003	$< 10^{-15}$
sample-pairs	LSB	low	−0.296	0.009	$< 10^{-15}$
sample-pairs	LSB	med	−0.109	0.008	$< 10^{-15}$
sample-pairs	LSB	high	−0.034	0.004	7.1×10^{-14}

E. Score-Level Paired t-Tests

Table 12 reports paired Student’s t-tests on the *raw detector scores* across the four pre-registered comparisons (3 source contrasts $\times$ 6 detectors + plain vs. AES-256-CBC $\times$ 6 detectors = 24 tests). Holm correction is applied within the 24-test family. Effect sizes are Cohen’s d (small=0.2, medium=0.5, large=0.8). The encryption comparisons (top block) all have $|d| \leq 0.029$ and are consistent with the RQ5 null; the carrier-source comparisons show large effects on the strongest detectors (RS, sample-pairs) and the new χ^2 -DCT-tiled.

Reading the t-test against DeLong. The SDXL-vs-FLUX comparisons all show large score-level effects ($|d|=0.06$ to 0.49) yet collapse to $|\Delta_{\text{AUC}}| < 0.05$ at every stratum under DeLong. This is the textbook pattern of *score-shift without ranking-shift*: detectors see the two diffusion models as distinct *distributions* but cannot exploit the difference for stego-vs-cover ranking. The RQ5 (plain vs AES) row shows the opposite for the spatial detectors (small but Holm-significant t-test, AUC null), which we interpret in the discussion as a deterministic-payload-content artefact below the AUC noise floor.

F. Score-Level Paired Wilcoxon Tests

Table 13 reports the non-parametric Wilcoxon signed-rank counterpart of the t-tests, with rank-biserial effect size r and Holm correction within the same 24-test family. Conclusions agree with the t-tests; we include the Wilcoxon results because chi-square detector scores have heavy right tails and the t-test’s normality assumption is mildly violated on those detectors.

TABLE 12

SCORE-LEVEL PAIRED T-TESTS ACROSS ALL PRE-REGISTERED COMPARISONS. n IS THE NUMBER OF PAIRED SCORE SAMPLES PER TEST.

Comparison	Detector	n	t	p	p_{Holm}	Cohen's d	Sig.
plain vs AES	cal- χ^2	27 000	-0.57	0.57	1.00	-0.003	
plain vs AES	χ^2 -DCT	27 000	-0.77	0.44	1.00	-0.005	
plain vs AES	χ^2 -DCT-tiled	27 000	+0.12	0.90	1.00	+0.001	
plain vs AES	χ^2 -spatial	27 000	+4.82	10^{-6}	3.5×10^{-5}	+0.029	✓
plain vs AES	RS	27 000	+4.16	3.2×10^{-5}	7.6×10^{-4}	+0.025	✓
plain vs AES	sample-pairs	27 000	+4.33	1.5×10^{-5}	3.6×10^{-4}	+0.026	✓
real vs SDXL	cal- χ^2	18 000	+9.66	$< 10^{-15}$	$< 10^{-15}$	+0.072	✓
real vs SDXL	χ^2 -DCT	18 000	-19.97	$< 10^{-15}$	$< 10^{-15}$	-0.149	✓
real vs SDXL	χ^2 -DCT-tiled	18 000	-22.17	$< 10^{-15}$	$< 10^{-15}$	-0.165	✓
real vs SDXL	χ^2 -spatial	18 000	+15.77	$< 10^{-15}$	$< 10^{-15}$	+0.118	✓
real vs SDXL	RS	18 000	-51.69	$< 10^{-15}$	$< 10^{-15}$	-0.385	✓
real vs SDXL	sample-pairs	18 000	+53.79	$< 10^{-15}$	$< 10^{-15}$	+0.401	✓
real vs FLUX	cal- χ^2	18 000	-15.52	$< 10^{-15}$	$< 10^{-15}$	-0.116	✓
real vs FLUX	χ^2 -DCT	18 000	-34.71	$< 10^{-15}$	$< 10^{-15}$	-0.259	✓
real vs FLUX	χ^2 -DCT-tiled	18 000	-48.67	$< 10^{-15}$	$< 10^{-15}$	-0.363	✓
real vs FLUX	χ^2 -spatial	18 000	+9.14	$< 10^{-15}$	$< 10^{-15}$	+0.068	✓
real vs FLUX	RS	18 000	-71.49	$< 10^{-15}$	$< 10^{-15}$	-0.533	✓
real vs FLUX	sample-pairs	18 000	+34.80	$< 10^{-15}$	$< 10^{-15}$	+0.259	✓
SDXL vs FLUX	cal- χ^2	18 000	-65.19	$< 10^{-15}$	$< 10^{-15}$	-0.486	✓
SDXL vs FLUX	χ^2 -DCT	18 000	-58.97	$< 10^{-15}$	$< 10^{-15}$	-0.440	✓
SDXL vs FLUX	χ^2 -DCT-tiled	18 000	-70.68	$< 10^{-15}$	$< 10^{-15}$	-0.527	✓
SDXL vs FLUX	χ^2 -spatial	18 000	-8.06	$< 10^{-15}$	$< 10^{-15}$	-0.060	✓
SDXL vs FLUX	RS	18 000	-27.53	$< 10^{-15}$	$< 10^{-15}$	-0.205	✓
SDXL vs FLUX	sample-pairs	18 000	-24.99	$< 10^{-15}$	$< 10^{-15}$	-0.186	✓

TABLE 13

SCORE-LEVEL PAIRED WILCOXON SIGNED-RANK TESTS ACROSS ALL PRE-REGISTERED COMPARISONS.

Comparison	Detector	n	W	p	p_{Holm}	r	Sig.
plain vs AES	cal- χ^2	27 000	1.82×10^8	0.69	1.00	-0.003	
plain vs AES	χ^2 -DCT	27 000	1.81×10^8	0.44	1.00	-0.005	
plain vs AES	χ^2 -DCT-tiled	27 000	1.68×10^8	0.79	1.00	-0.002	
plain vs AES	χ^2 -spatial	27 000	1.77×10^8	1.3×10^{-5}	3.2×10^{-4}	+0.031	✓
plain vs AES	RS	27 000	1.76×10^8	2.7×10^{-5}	6.5×10^{-4}	+0.030	✓
plain vs AES	sample-pairs	27 000	1.68×10^8	$< 10^{-15}$	$< 10^{-15}$	+0.072	✓

All carrier-source pairs (18 rows) are Holm-significant at $p_{\text{Holm}} < 10^{-15}$;
effect sizes track Cohen's d from the t -test table (small r on chi-square detectors,
large r on RS / sample-pairs / χ^2 -DCT-tiled). See `wilcoxon_tests.csv` for the full 24-row listing.

G. Cover-only PoV Imbalance Diagnostic

Table 14 reports the full per-source distribution of the cover-only PoV imbalance S from Equation 1 ($n=3,000$ covers per source, no embedding) and the paired Wilcoxon signed-rank tests against the caption-matched real baseline. The same data drives Figure 11 in Section F.

The signed differences are both negative: the naive hypothesis (real $>$ ML) is rejected with overwhelming significance. ML covers have *slightly higher* median PoV imbalance than real covers, with paired effect $|d_z|=0.12$. The variance is also higher on ML covers (std 11.9×10^3 real vs 18.2×10^3 ml_a vs 14.1×10^3 ml_b), and the cover-side χ^2 -spatial score variance follows the same ordering (std 37.5, 54.4, 43.0). See Section F for the corrected mechanism.

H. Reproducibility

The full pipeline is open source, seeded at every stage (split, payload, embedding, ML generation), and idempotent (re-runs skip work already on disk). Every run lives in `runs/<id>/` with a frozen `config.json`; the figure and table numbers in Section V can be regenerated from `predictions.csv` alone. The repository contains 245 automated tests covering detectors, embedders, RQ verdict logic, and the analysis pipeline. The supplementary partial-AUC analysis is run separately via `scripts/compute_pauc_analysis.py`, which uses the same predictions as the primary analysis.

TABLE 14
COVER-ONLY PoV IMBALANCE $S = \sum_k |n_{2k} - n_{2k+1}|$ PER CARRIER SOURCE. PAIRED WILCOXON TESTS ARE AGAINST THE
CAPTION-MATCHED REAL COVER.

Source	Q1	median	Q3	mean	std
real	7,418	9,698	14,051	13,003	11,945
ml_a	8,181	10,272	14,520	15,532	18,166
ml_b	8,644	11,649	16,424	15,003	14,128

Paired Wilcoxon (signed-rank), one-sided real > ML:
real vs ml_a $W = 1.95 \times 10^6$, $p = 4.0 \times 10^{-10}$, median diff = -653
real vs ml_b $W = 1.69 \times 10^6$, $p = 3.6 \times 10^{-32}$, median diff = -1,518